NTI DATA ANALYSIS FINAL PROJECT

RAW DATA TO ACTIONABLE INSIGHTS

AUGEST 2026

National Telecommunication Institute

OUR TEAM

01 **Hla Hany**
Exploring The Data

02 **Mohammed Atef**
Data Processing And Cleaning

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Answering Bussiness Questions

04 **Youssef Aly**
Dashboard And Reporting

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IBM

IBM

IBM

01

BUSINESS PROBLEM & OBJECTIVES

Our Team act as People Analytics analysts for a IBM's HR department, investigating why employees leave and which departments/roles are most affected, to inform retention strategy.

WORKFLOW

EXCEL 1 → PYTHON 2 → SQL 3 → POWER BI 4

01

BUSINESS PROBLEM & OBJECTIVES

Exploring & Cleaning The Data

Understanding the data we are dealing with, exploring dataset inconsistencies, null values, duplicate data, and the kinds of columns we are dealing with, an then cleaning it.

Answering Bussiness Questions

After Cleaning the Dataset , We go on exploring what the data have to offer by answering bussiness questions, discovering patterns And Extracting Important insights

Dashboarding & Reporting

Now that we have explored the dataset and extracted important insights, we can move on to visualizing the data for non-technical people, making the insights easier to understand and communicate.

02

ABOUT THE DATA

Dataset Name

- IBM HR Analytics Employee Attrition & Performance

Dataset Source

- https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Dataset Format

- CSV

02

ABOUT THE DATA (DATASET AT A GLANCE)

1470
Employee Records

35
Columns

3
One Value Columns

CORE METRICS / COLUMNS

Attrition
- YES
- NO

Depart.
- SALES
- HUMAN RECOURCES
- RESEARCH & DEVELOPMENT

Marital Statues
- SINGLE
- MARRIED
- DIVORCED

Job Role
- HEALTHCARE REPRESENT.
- HUMAN RESOURCES
- LABORATORY TECHNICIAN
- MANAGER
- MANUFACTURING DIRECTOR
- RESEARCH DIRECTOR
- RESEARCH SCIENTIST
- SALES EXECUTIVE
- SALES REPRESENTATIVE

EmployeeCount
- 1 ONLY

Over 18
- Y ONLY

StandardHours
- 80 ONLY

02

ABOUT THE DATA (DATASET AT A GLANCE)

Important KPIs

Total Employees

1,470

Attrition Count

237

Attrition Rate

16.12%

Avg. Monthly Income

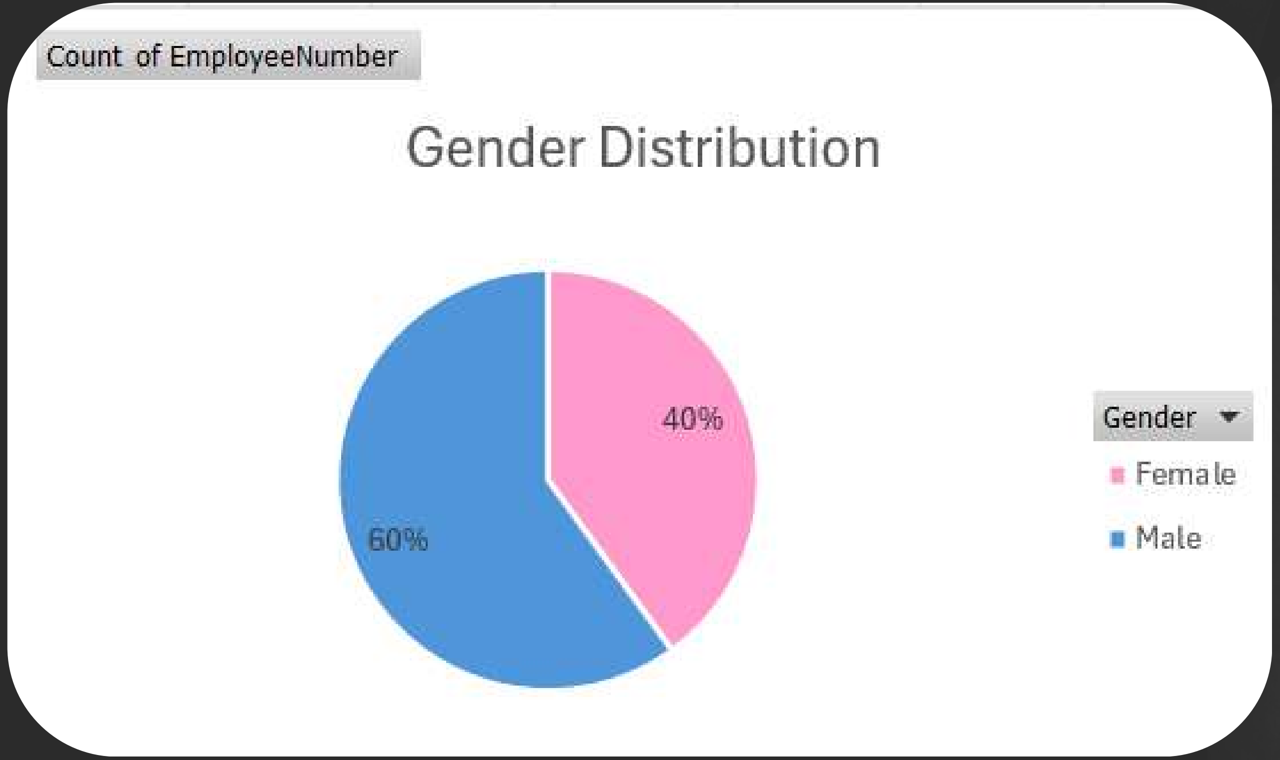
6502.93

Avg. Job Satisfaction

2.73

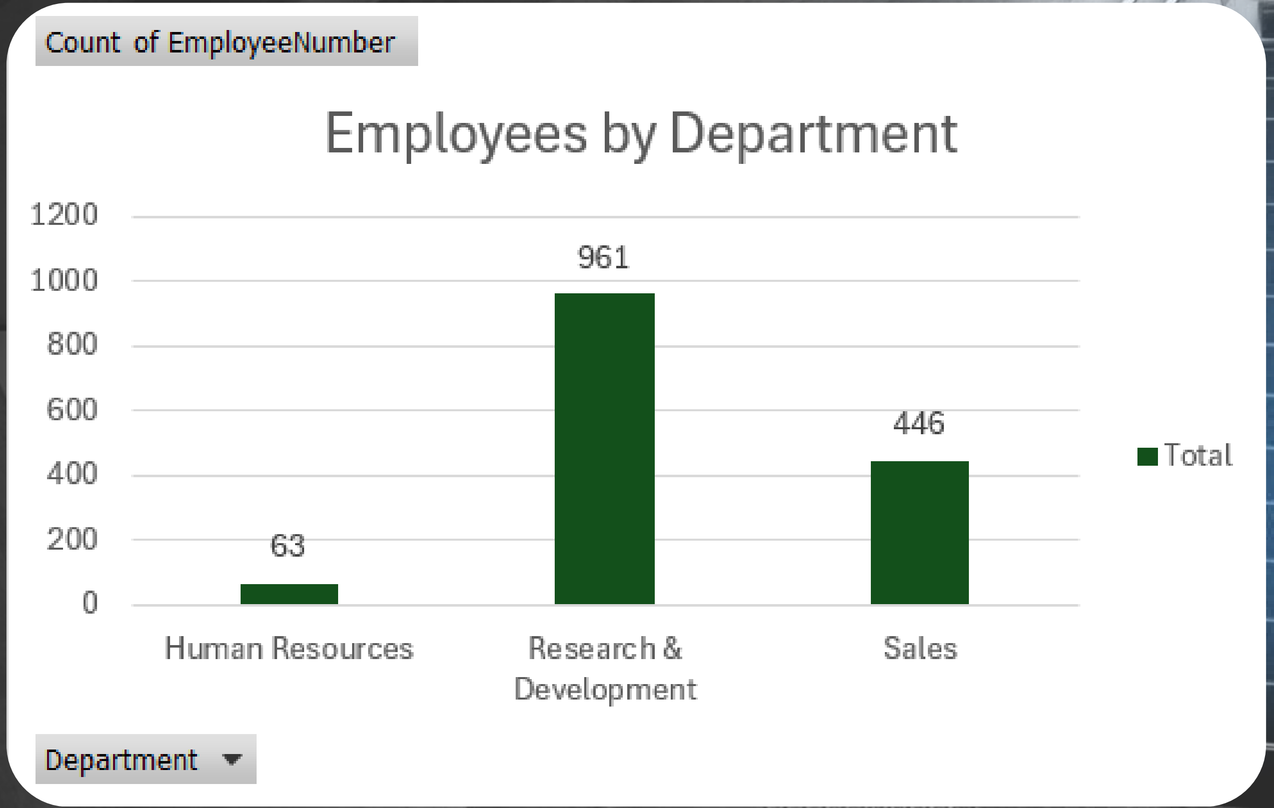
1

How Are Genders Distributed Across the Dataset?



2

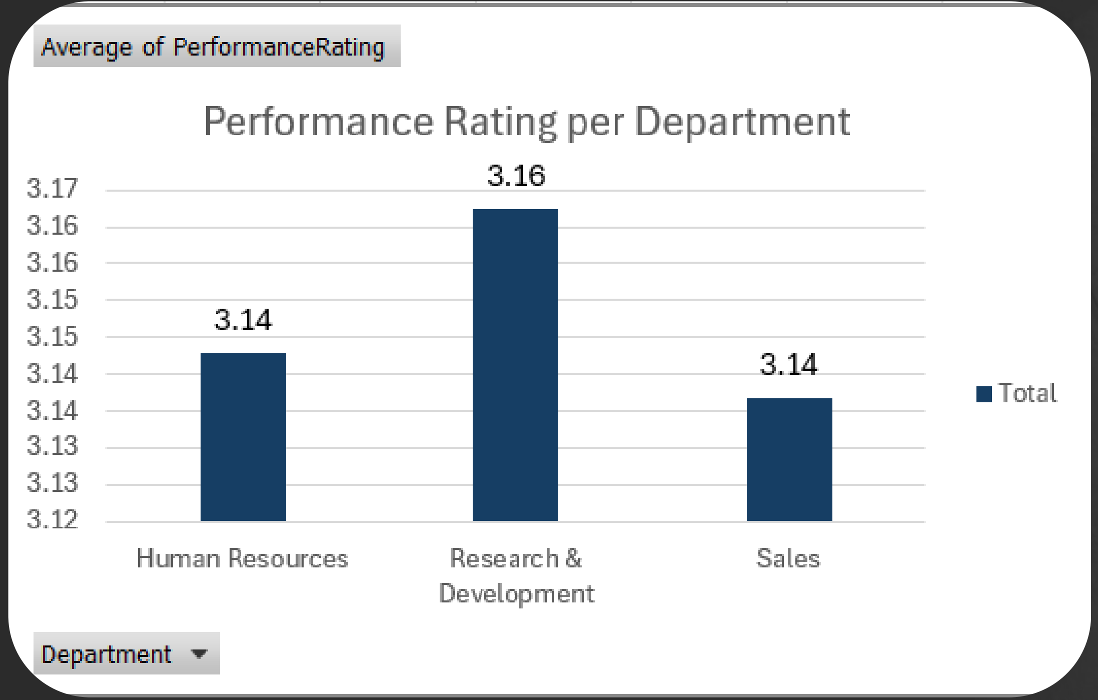
How Are Employees Distributed Across Departments?



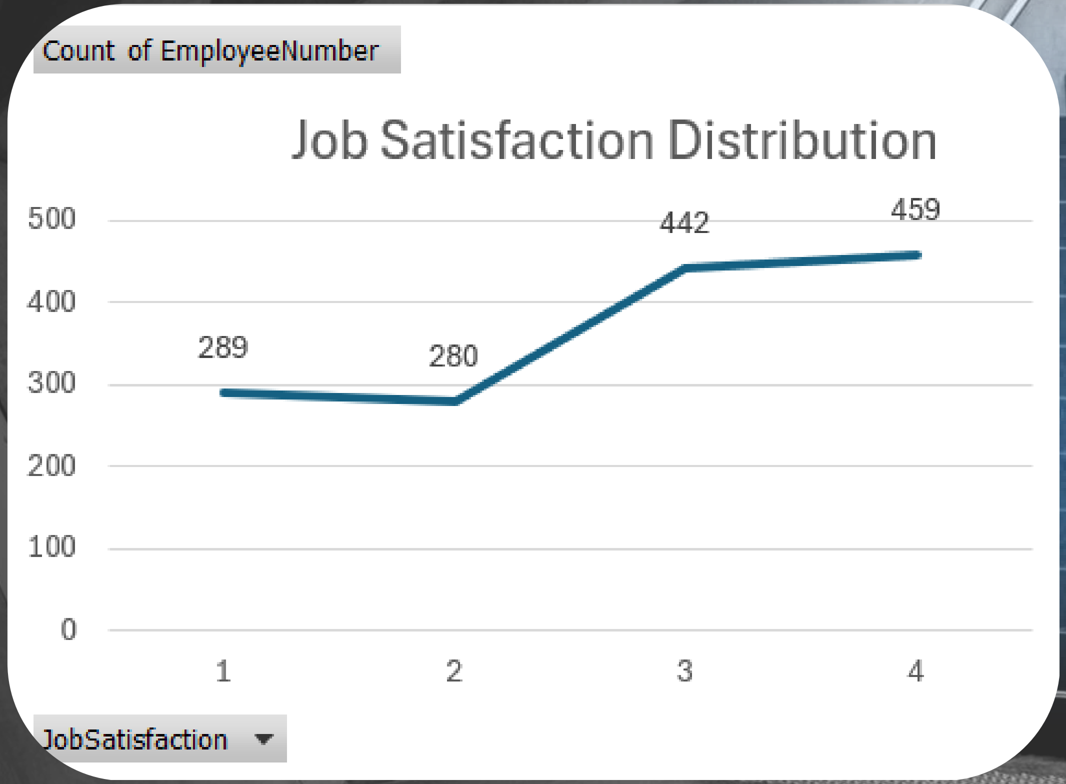
02
ABOUT THE DATA (DATASET AT A GLANCE)

3
What Is the Attrition Rate Distribution?

4
What Is the Preformance Rating Per Department??



5
What is the distribution of job satisfaction levels among employees?



03
DATA CLEANING AND PROCESSING

1 No Duplicates Detected

2 No Nulls Found

```
print(f"Dataset Shape: {dataset.shape}")
print(f"Duplicates Count: {dataset.duplicated().sum()}")
print(f"Missing Values Total: {dataset.isna().sum().sum()}")
```

```
Dataset Shape: (1470, 35)
Duplicates Count: 0
Missing Values Total: 0
```

03
DATA CLEANING AND PROCESSING

3 Dropped 4 Columns
*All containing One Value

```
constant_cols = [col for col in dataset.columns if
dataset[col].nunique() <= 1]
cols_to_drop = constant_cols + ["EmployeeNumber"]
print(f"Columns to drop: {cols_to_drop}")

# Output
Columns to drop:
['EmployeeCount', 'Over18', 'StandardHours', 'EmployeeNumber']
```

04

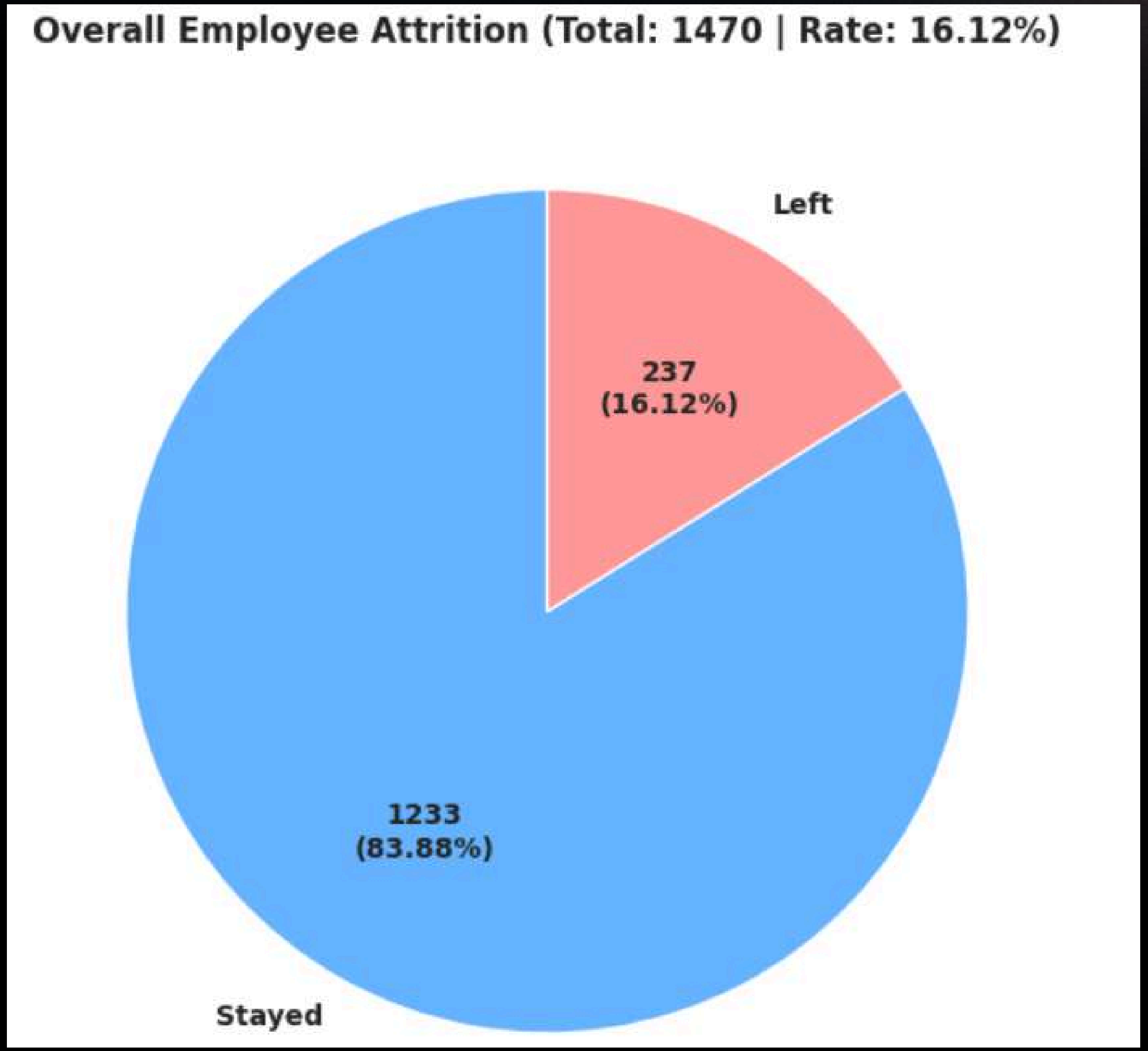
EXPLORATORY ANALYSIS

(ANSWERING BUSINESS QUESTIONS)

Understand overall attrition rate and how it varies across departments

```
select
    count(*) as Total_Emps,
    sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,
    sum(case when Attrition = 0 then 1 else 0 end) as Emps_Stayed,
    cast(
        sum(case when Attrition = 1 then 1 else 0 end) * 100.0
        / count(*)
        as decimal(5,2)
    ) as Attrition_Rate
from Employees;
```

	Total_Emps	Emps_Left	Emps_Stayed	Attrition_Rate
1	1470	237	1233	16.12



04

EXPLORATORY ANALYSIS

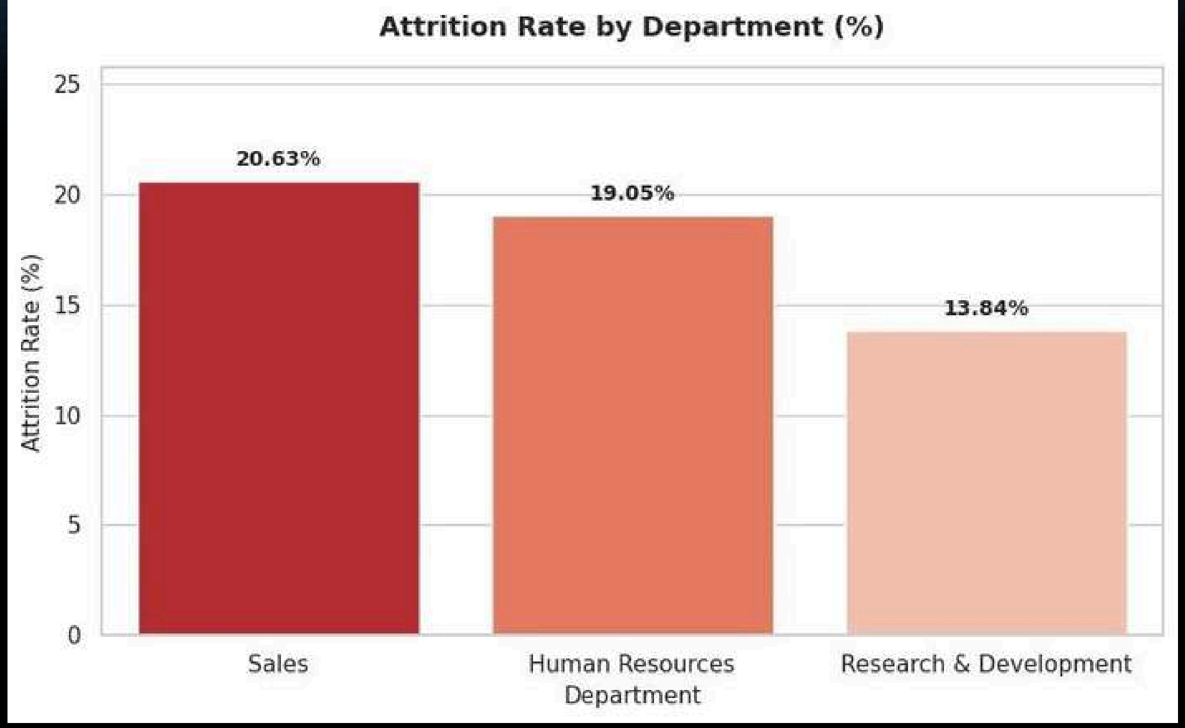
(ANSWERING BUSINESS QUESTIONS)

What is the attrition rate by department, job role, and overtime status?

DEPARTMENT

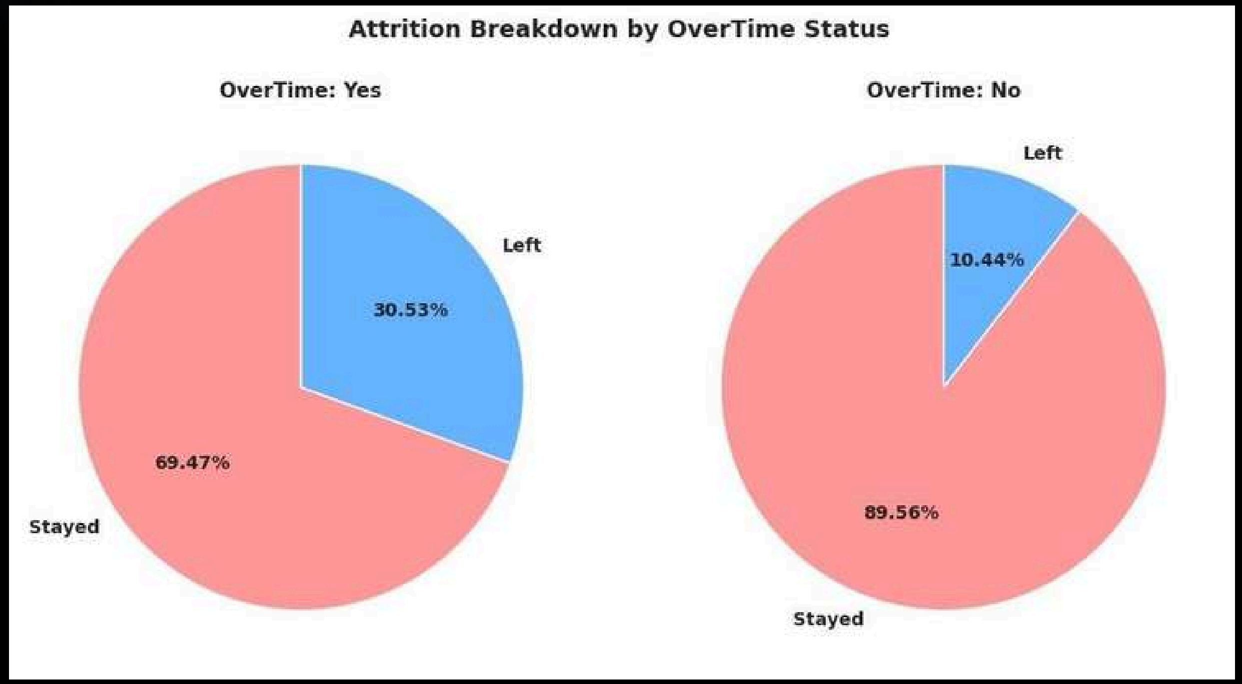
```sql
select
    Department,
    count(*) as Total_Emps,
    sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,

     cast(
     sum(case when Attrition = 1 then 1 else 0 end) * 100.0
     / count(*)
     as decimal(5,2)
     ) as Attrition_Rate
from Employees
group by Department
order by Attrition_Rate desc;
```



OVERTIME

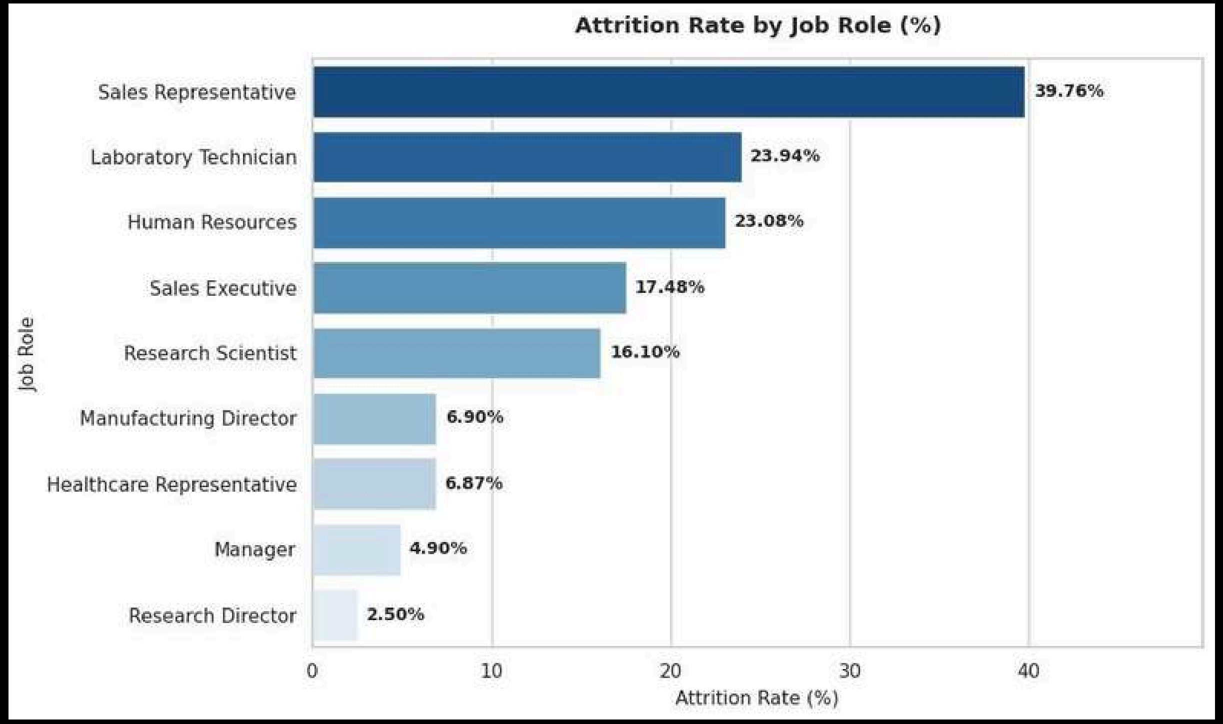
```sql
select
    case
        when OverTime = 1 then 'Yes'
        else 'No'
    end as OverTime_Status,
    count(*) as Total_Emps,
    sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,
    cast(sum(case when Attrition = 1 then 1 else 0 end) * 100.0/
count(*)as decimal(5,2)
     ) as Attrition_Rate
from Employees
group by OverTime
order by Attrition_Rate desc;
```



JOB ROLE

```sql
select
    JobRole,
       count(*) as Total_Emps,
       sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,

       cast(
       sum(case when Attrition = 1 then 1 else 0 end) * 100.0
       / count(*)
       as decimal(5,2)
        ) as Attrition_Rate
from Employees
group by JobRole
order by Attrition_Rate desc;
```



04

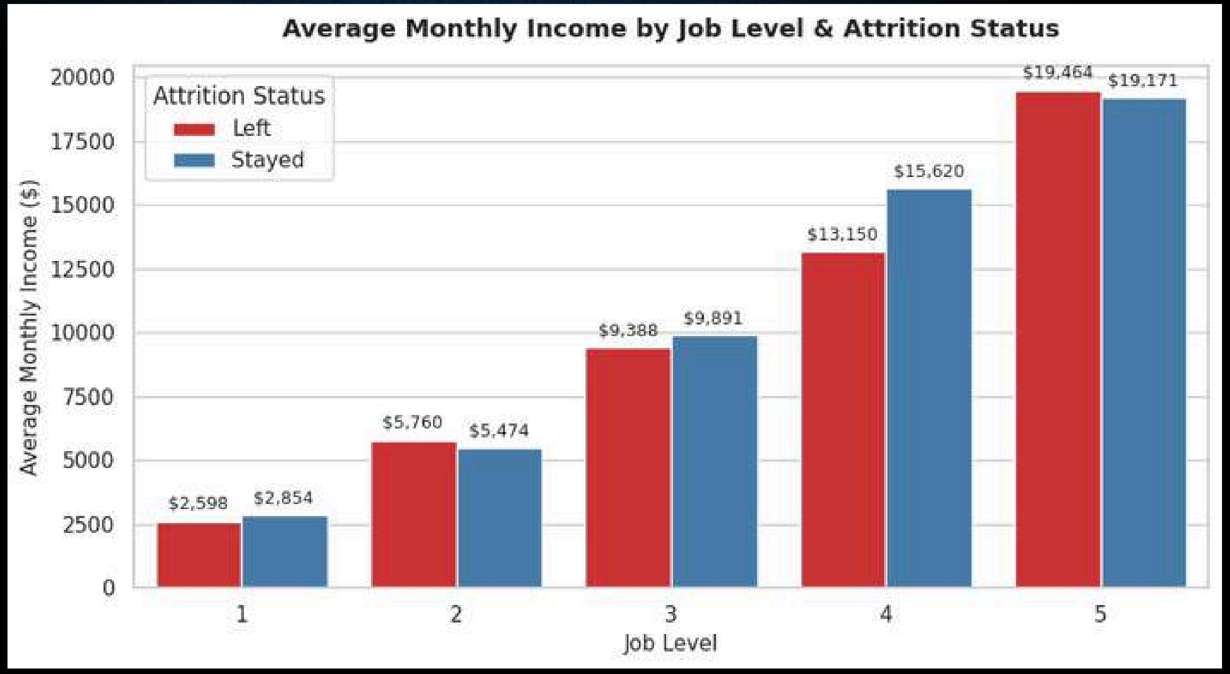
EXPLORATORY ANALYSIS

(ANSWERING BUSINESS QUESTIONS)

What is the average monthly income by job level and attrition status?

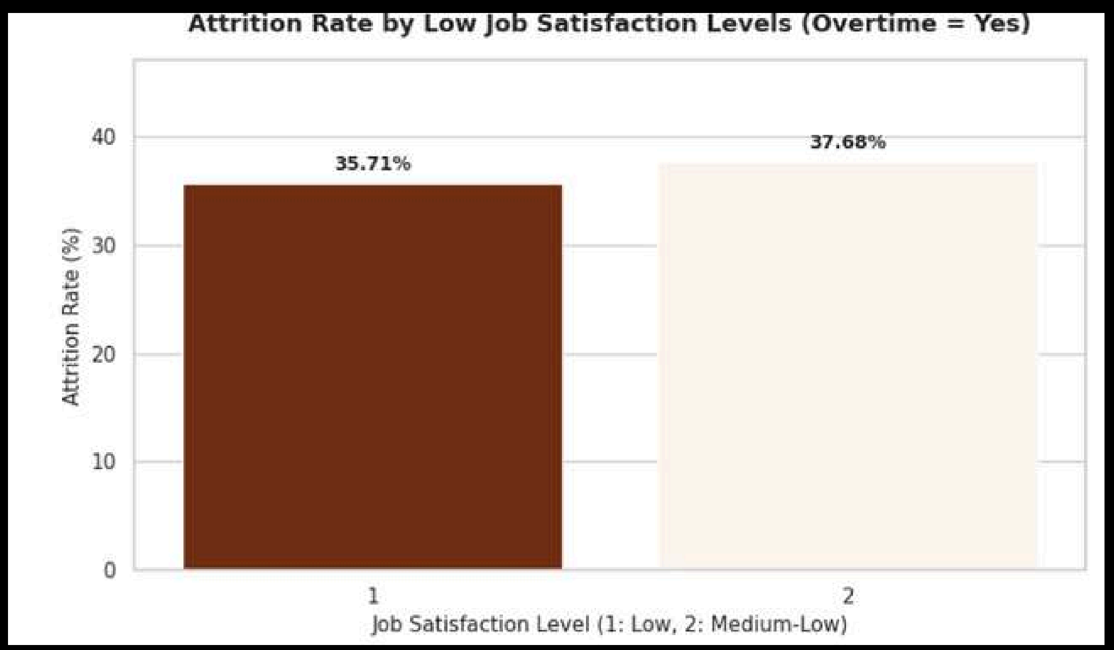
```
select
    JobLevel,
    case
        when Attrition = 1 then 'Left'
        else 'Stayed'
    end as AttritionStatus,

    cast(avg(MonthlyIncome) as decimal(10,2)) as Avg_MonthlyIncome
from Employees
group by
    JobLevel,
    Attrition
order by
    JobLevel,
    AttritionStatus;
```



Which employees combine low job satisfaction with frequent overtime?

```
select
    JobSatisfaction,
    case
        when OverTime = 1 then 'Yes'
        else 'No'
    end as OverTimeStatus,
    count(*) as Total_Emps,
    sum(case when Attrition = 1 then 1 else 0 end) as Emps_left,
    cast(
        sum(case when Attrition = 1 then 1 else 0 end) * 100.0
        / count(*)
        as decimal(5,2)
    ) as AttritionRate
from Employees
where JobSatisfaction <= 2
and OverTime = 1
group by
        JobSatisfaction,
        OverTime
order by
        JobSatisfaction;
```

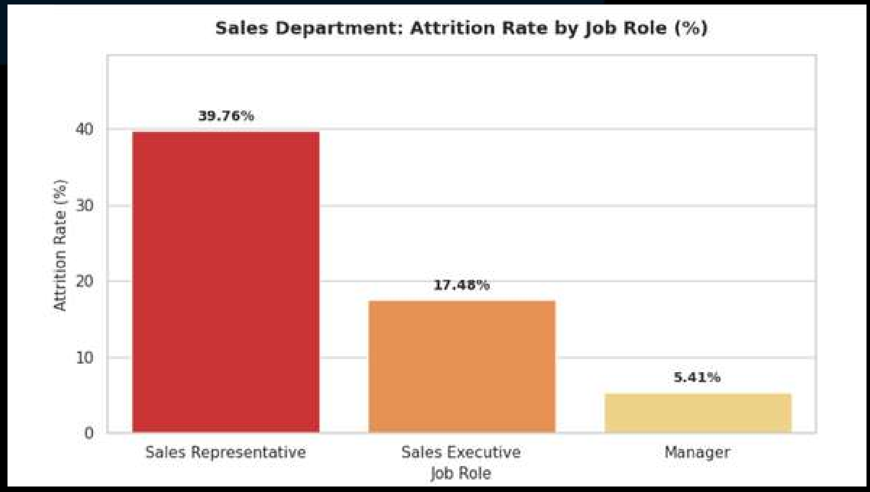


Which department has the highest attrition, and how does it break down by job role within that department? + Diagonstic

```
select top 1
    Department,
    cast(
        sum(case when Attrition = 1 then 1 else 0 end) * 100
        / count(*)
        as decimal(5,2)
        ) as AttritionRate
from Employees
group by Department
order by AttritionRate desc;
```

	Department	AttritionRate
1	Sales	20.63

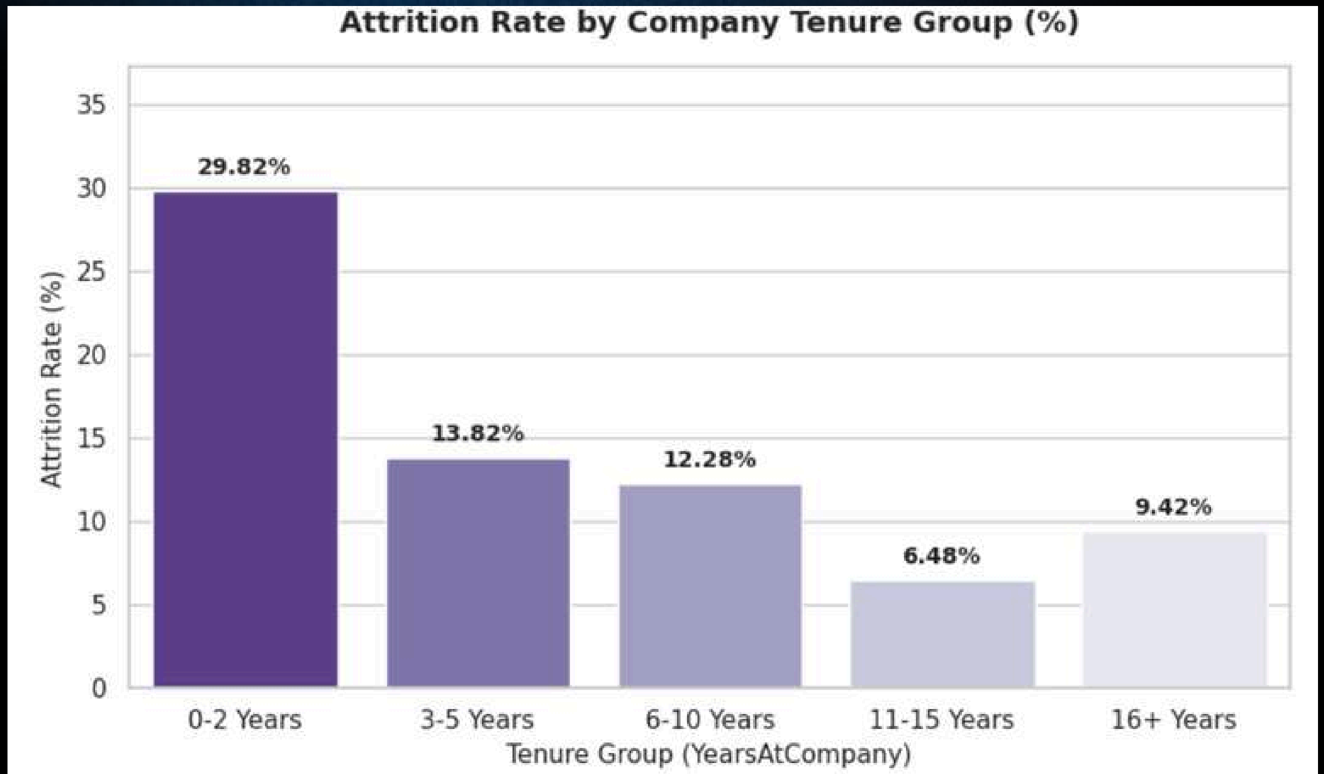
```
select
    JobRole,
    cast(
        sum(case when Attrition = 1 then 1 else 0 end) * 100.0
        / count(*)
        as decimal(5,2)
        ) as AttritionRate
from Employees
where Department = 'Sales'
group by JobRole
order by AttritionRate desc;
```



04
EXPLORATORY ANALYSIS
(DIAGNOSTIC ANALYTICS QUESTIONS)

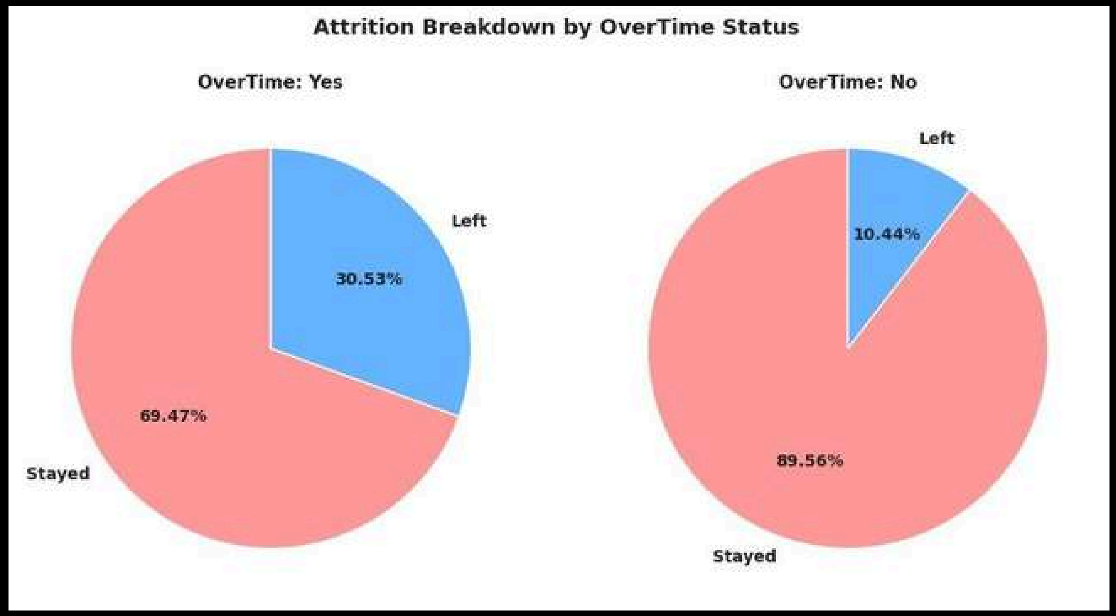
Bucket "years at company" into tenure groups and compute attrition rate per group

```sql
select
    case
        when YearsAtCompany <= 2 then '0-2 Years'
        when YearsAtCompany <= 5 then '3-5 Years'
        when YearsAtCompany <= 10 then '6-10 Years'
        when YearsAtCompany <= 15 then '11-15 Years'
        else '16+ Years'
    end as TenureGroup,
    cast(
        sum(case when Attrition = 1 then 1 else 0 end) * 100.0
        / count(*)
        as decimal(5,2)
        ) as AttritionRate
from Employees
group by
    case
        when YearsAtCompany <= 2 then '0-2 Years'
        when YearsAtCompany <= 5 then '3-5 Years'
        when YearsAtCompany <= 10 then '6-10 Years'
        when YearsAtCompany <= 15 then '11-15 Years'
        else '16+ Years'
    end;
```



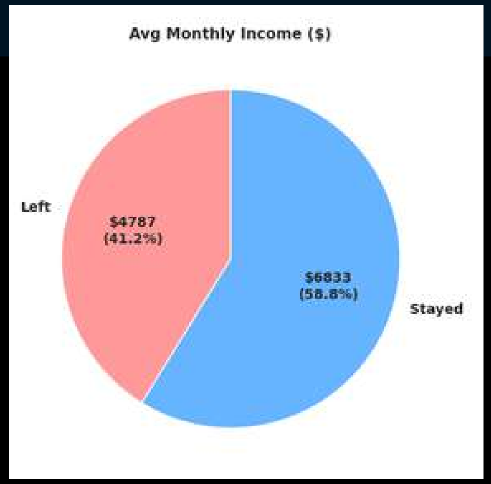
Cross-tabulate overtime status against attrition, and job satisfaction score against attrition

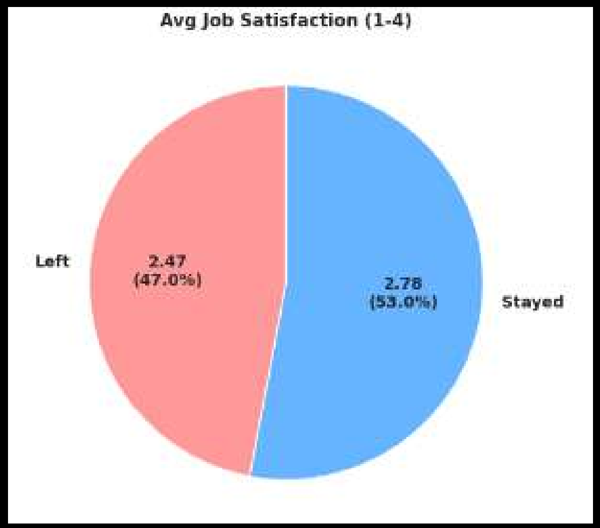
```sql
select
    case
        when OverTime = 1 then 'Yes'
        else 'No'
    end as OverTime_Status,
    sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,
    sum(case when Attrition = 0 then 1 else 0 end) as Emps_Stayed
from Employees
group by OverTime;
```

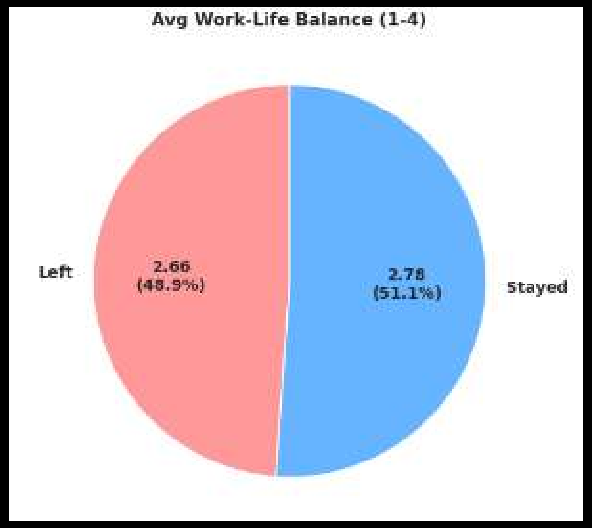


Compare average income, satisfaction, and work-life balance scores between employees who left vs. stayed

```sql
select
    case
        when Attrition = 1 then 'Left'
        else 'Stayed'
    end as AttritionStatus,
    avg(MonthlyIncome) as Avg_Income,
    avg(JobSatisfaction) as Avg_Satisfaction,
    avg(WorkLifeBalance) as Avg_WorkLife_Balance
from Employees
group by Attrition;
```







05 DASHBOARDING & REPORTS

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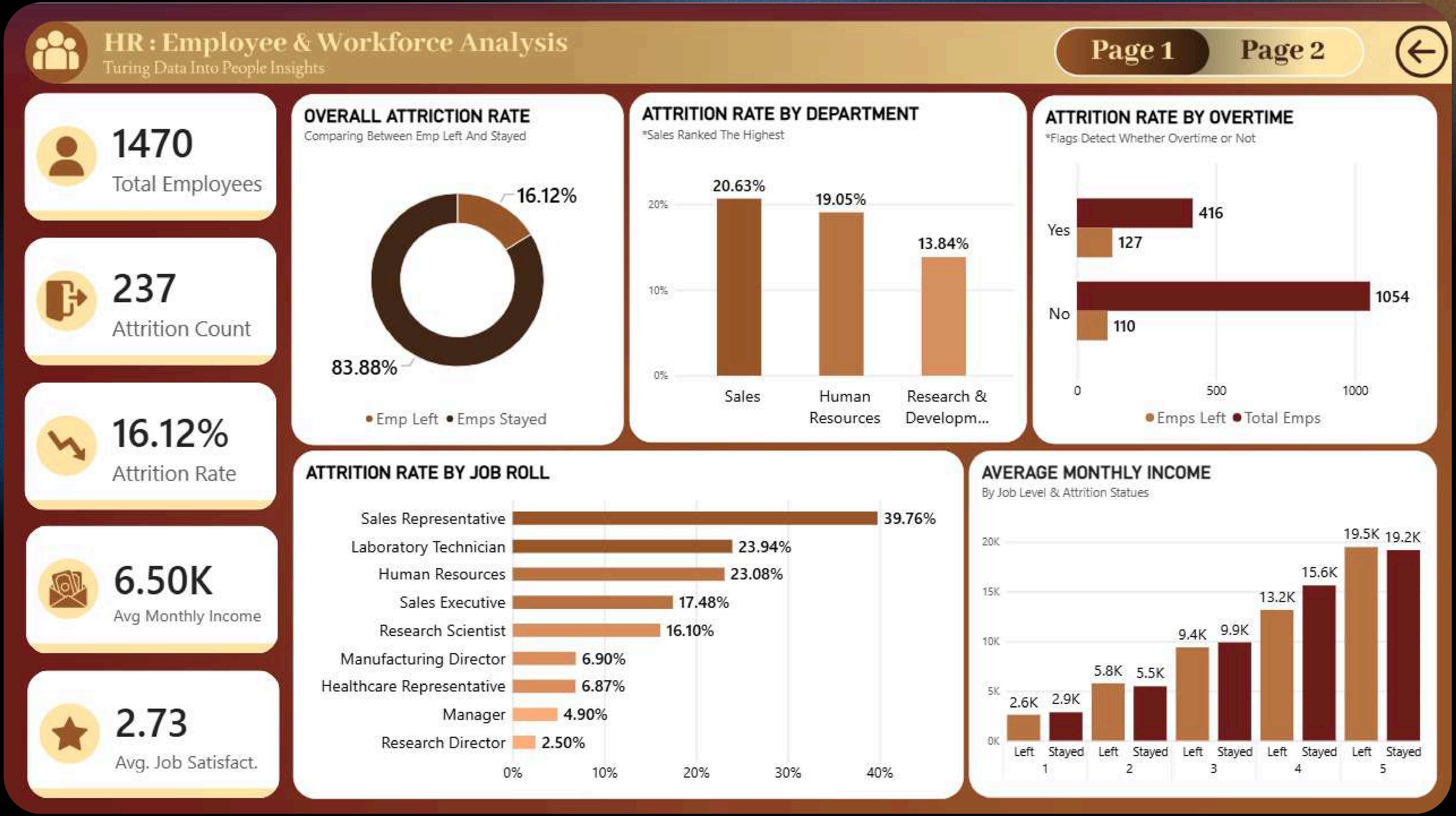
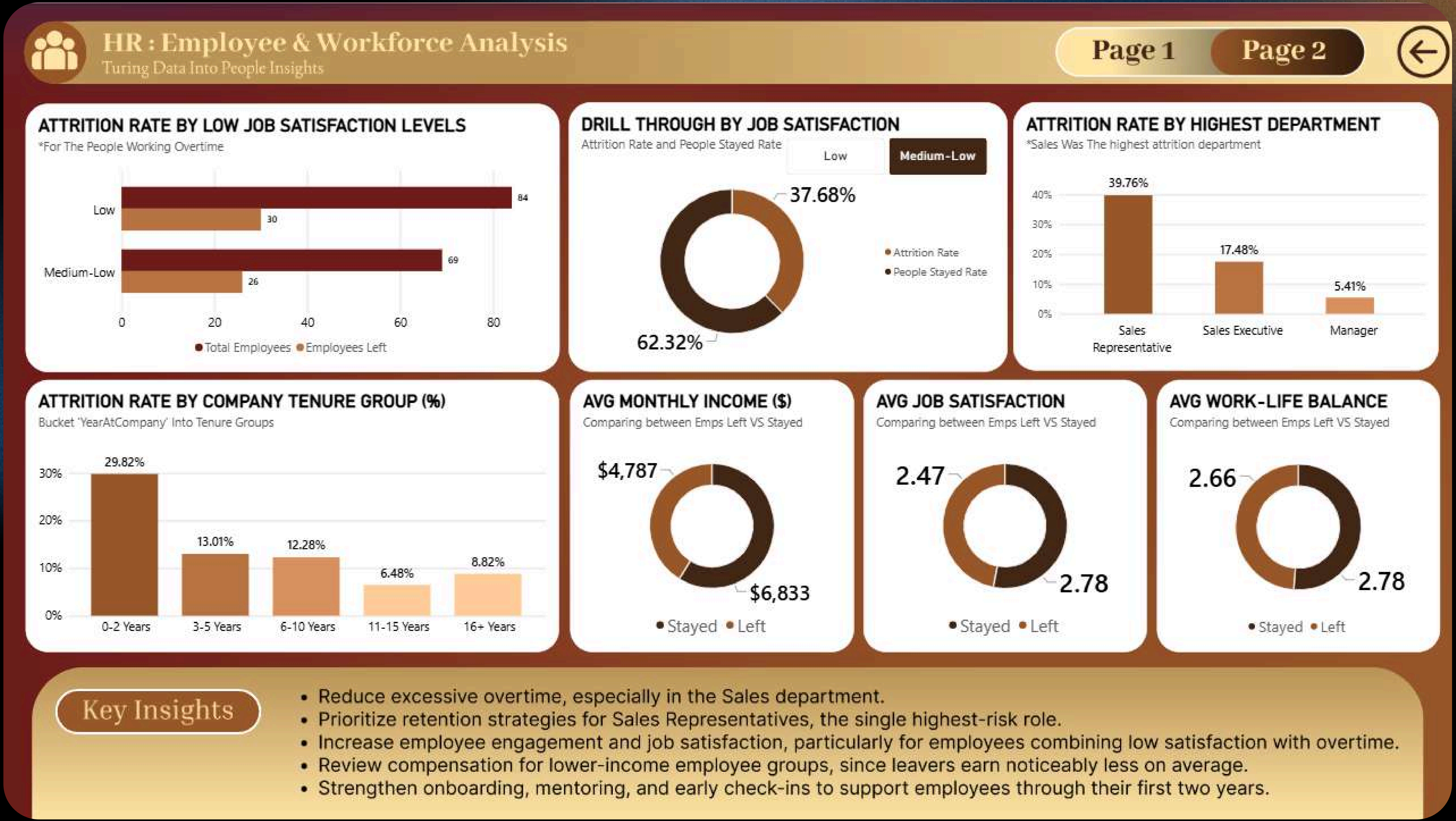


05 DASHBOARDING & REPORTS



THANK YOU!